Open-ended agentic search for neural encoding models

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Abstract Visual encoding models typically inherit representations optimized for computer-vision objectives, leaving neural data to evaluate models rather than determine how they are constructed. We introduce neuro-autoresearch, which uses held-out neural predictivity to search directly over executable visual programs. Agents revise candidate code, while a frozen evaluator applies the same dataset-specific ridge-readout procedure and returns a scalar score for each proposal. Candidate parameters remain fixed during evaluation, and candidate code cannot access neural targets. We perform independent searches in human 7 T functional magnetic resonance imaging (NSD) and two macaque electrophysiology datasets (Triple-N and TVSD). Each search selects a distinct program composed of explicit operations such as multiscale filtering, color opponency, normalization and pooled form and texture statistics. In the predefined search cohorts, the selected programs score above all measured references in early visual regions but not in later ventral or inferotemporal cortex. The same regional ordering appears in individuals excluded from model selection. Neural-guided program search therefore discovers fixed-weight representations with an early-visual advantage without beginning from a pretrained computer-vision model.

Introduction

A central aim of visual neuroscience is to identify the computations that transform images into neural responses. Encoding models make this question experimentally tractable: a model maps a stimulus to a feature representation, and a fitted readout tests whether those features account for measured activity.[1] Predictivity alone, however, does not identify why a representation succeeds. A model may capture a relevant computation, provide a broad basis that a linear readout can exploit, or inherit useful structure from its training history. The scientific value of an encoding model therefore depends not only on its score, but also on how the candidate representation was constructed and what alternatives were considered.[2]

Modern visual encoding research has benefited greatly from artificial networks developed for computer vision and evaluated against neural data.[3–8] Supervised and self-supervised systems provide rich representations and useful reference points for the biological visual hierarchy. Yet their defining objective is an engineering task such as classification, image-text alignment or representation learning. Neural predictivity is usually measured after training rather than used to determine which visual function should be built. As these systems grow, their representations also combine the effects of architecture, data, objective functions, optimization and many learned parameters. Restricting model comparison to this inherited set leaves much of the space of possible encoding functions unexplored and complicates the use of deep networks as scientific models.[9,10]

Recent work from the Bonner laboratory sharpens both sides of this problem. Across diverse trained networks, brain alignment is concentrated in a small set of universal representational dimensions.[21] At the same time, human visual cortex expresses reliable structure across a much broader, scale-free spectrum whose dimensions are substantially shared across individuals.[22] Separately, cortex-aligned representations can emerge in sufficiently wide convolutional architectures even before task training, showing that architectural inductive bias can itself be consequential.[23] Together, these findings motivate a search

framework that can test both learned and untrained sources of neural alignment without prespecifying a single model family.

We introduce neuro-autoresearch, an agent-guided search over executable encoding models. The search is open-ended in a specific sense: agents can alter the sequence of operations and the structure of the candidate program rather than selecting only among members of one predefined architecture family. For each proposal, a frozen evaluator fits a common neural readout and returns predictivity on held-out stimuli. The agent sees this score and uses it to propose subsequent programs, so neural responses guide the search even though candidate code cannot read neural targets and its parameters are not optimized by gradient descent.

The method is not assumption-free. The program interface, permitted operations, agent proposal process, neural readout and reward together define the hypothesis space. Its central departure from the standard workflow is that the search need not begin from a representation pretrained on an external computer-vision objective. We implement the search with multiple independent model lineages and apply it separately to human fMRI and two macaque electrophysiology datasets. Each experiment evaluates whether the regional ordering of a model selected in a predefined search cohort appears again in individuals excluded from model selection. The study therefore tests model discovery and within-dataset individual holdout; it does not test direct transfer of one model across datasets or identify which program components are necessary neural mechanisms.

Methods

Search objective

For each candidate function f, the evaluator transforms a fixed stimulus set x into features f(x), fits ridge regression from those features to neural responses on a training split, and measures predictivity on held-out stimuli.[11] Within a dataset, every proposal uses the same stimulus split, feature-to-neural readout procedure and evaluation metric. Each candidate is therefore fixed during evaluation: the evaluator estimates only the linear ridge readout and does not update the parameters or operations of f. This design separates the representation being searched from the readout needed to compare that representation with recorded activity.

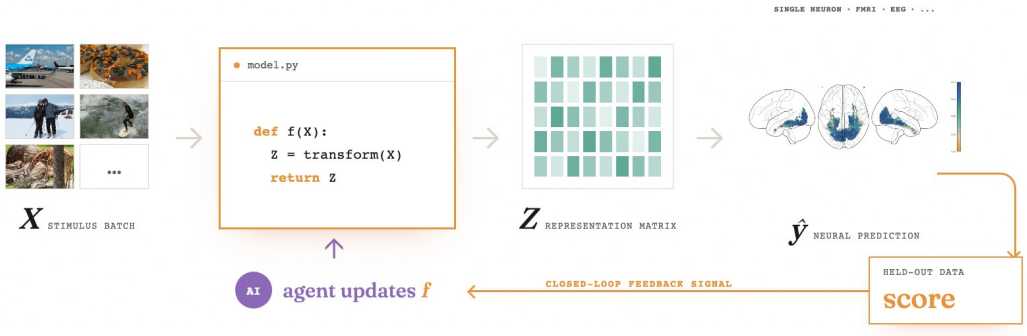


Figure 1 | Closed-loop search over visual encoding functions. A batch of natural images x is passed through an executable candidate function f to produce a feature matrix z. The frozen evaluator fits a dataset-specific ridge readout from z to recorded neural responses and scores the resulting predictions on held-out stimuli. Candidate parameters and operations remain fixed during this evaluation, and candidate code cannot access neural targets. The scalar predictivity score is returned to the agent, which edits f and submits a new candidate, closing the search loop.

Agentic model proposal and parallel search

Agents edit only the candidate model program. Each proposal receives image stimuli, must satisfy a fixed input-output interface and is executed by the frozen evaluator. It cannot load checkpoints, fetch models from a network or read neural-response files. Across iterations, however, the agent can change operations, constants, pooling schemes and program structure in response to previously returned scores. We use the term fixed-weight for an individual candidate and reserve search or optimization for this adaptive sequence of candidate programs.

To preserve multiple lines of search, independent islands maintain separate program ancestries. An island can continue from its own best program or inspect one randomly selected program from another island; the global winner is not broadcast as a common parent. A conductor reads the population and supplies broad advisory directions but does not control scores or admission to evaluation. Candidate programs enter an asynchronous queue, while shared database memory records code, scores, failures and ancestry (Fig. 2).

One 48-hour NSD implementation run used ten islands and successfully scored 3,599 candidate programs, with a mean evaluation time of 19.5 seconds. These values describe the operating scale of the search system, not the evidence for neural generalization. The scientific analyses below use predefined dataset-specific search cohorts, freeze the selected program, and evaluate it in individuals who did not contribute to its selection.

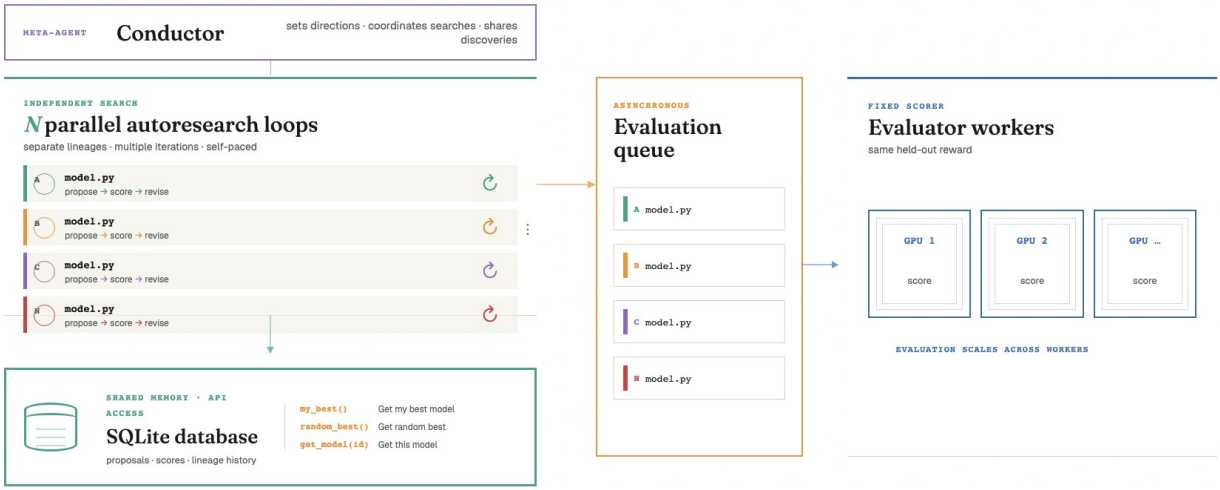


Figure 2 | Hierarchical and parallel search architecture. A conductor reads the shared search history and provides broad advisory directions to multiple independent search islands. Each island proposes, scores and revises its own model lineage, while a shared SQLite database stores candidate code, scores, failures and ancestry. Submitted programs enter an asynchronous evaluation queue and are distributed across fixed evaluator workers. All workers apply the same held-out neural objective, allowing model lineages to progress independently without changing the scoring procedure.

Datasets and individual holdout

We run a separate search for each dataset rather than selecting one model and transferring it across datasets (Table 1). NSD contains human responses to natural images measured with 7 T fMRI.[12] Subjects 1 and 2 define the search cohort, while subjects 5 and 7 are excluded from model selection. Triple-N contains dense macaque Neuropixels recordings during natural-image viewing.[13] Monkeys 1 to 3 define the search cohort and monkeys 4 and 5 define the held-out cohort. TVSD contains macaque spiking responses to THINGS images.[14] Monkey F defines the search cohort and monkey N is held out. Each dataset has its own selected program; no cross-dataset transfer is included in the reported comparisons.

Table 1 | Datasets, selection cohorts and evaluation metrics.

Dataset	Measurement	Model-selection cohort	Held-out evaluation
NSD	Human 7 T fMRI	Subjects 1 and 2	Subjects 5 and 7; noise-normalized explained variance
Triple-N	Macaque Neuropixels	Monkeys 1 to 3	Monkeys 4 and 5; noise-normalized predictivity
TVSD	Macaque spiking	Monkey F	Monkey N; mean Pearson correlation

Search and individual holdout are distinct from the stimulus split used to fit and test each neural readout. During search, candidate scores are computed only from the designated search cohort. After a program is selected, its code and fixed parameters are frozen. A new ridge readout is then fitted for each held-out individual using that individual's training stimuli and scored on that individual's test stimuli. Held-out neural responses therefore support the final readout fit and evaluation, but never influence which program is selected.

The main NSD analysis reports V1, V2, V3 and a ventral aggregate. Triple-N reports early visual cortex (EVC) and inferotemporal cortex (IT), and TVSD reports V1, V4 and IT. NSD and Triple-N use noise-normalized scores; TVSD reports mean Pearson correlation over reliable spiking units. Because the metrics, neural measurements and sampled populations differ, score magnitudes are interpreted only within a dataset.

Reference models and integrity controls

The searched encoders are compared with measured untrained and pretrained references under the same dataset-specific readout. The reference set includes ResNet-50, ViT-B/16, DINOv2 and CLIP visual encoders.[15–18] The evaluator, stimulus split, metric and readout remain frozen during search. An audit of 3,604 submitted programs from the featured NSD run finds no checkpoint loading, network fetching, gradient training, neural-target leakage or direct dataset reads inside candidate models. This establishes the integrity of candidate execution. It does not make the comparison symmetric: the searched encoder is selected from thousands of proposals, whereas each plotted pretrained reference is one prespecified model.

Results

We first describe the programs selected by the three searches, then evaluate their regional predictivity. The NSD experiment tests human fMRI; Triple-N tests macaque single-unit responses measured with Neuropixels; and TVSD uses a different macaque image collection and response metric. In each dataset, the central comparison is descriptive: does the regional ordering measured in the search cohort appear again in individuals excluded from model selection?

Search produces explicit fixed-weight visual programs

The selected models are source-level programs rather than only identifiers for learned checkpoints. The NSD program combines multiscale luminance-oriented energy, red-green and blue-yellow opponent energy, center-surround filtering, divisive normalization and second-order filtering of oriented-energy maps. It pools these responses into retinotopic, spectral, color-composition and texture statistics. Its features are computed from analytic or fixed-weight operations; only the downstream ridge readout is fitted to neural responses.

The independently selected macaque programs use overlapping visual primitives but are not copies of the NSD model. The Triple-N program includes opponent-color signals, multiscale oriented energy, contrast normalization, global spectral summaries and non-Cartesian form descriptors. The TVSD

program combines a five-scale anisotropic Gabor bank with spatial and color statistics, orientation co-occurrence, cross-scale products, local texture summaries and a fixed hierarchical form pathway. These descriptions identify what the selected programs compute; they do not show that any named component is necessary, sufficient or implemented by the biological visual system.

Direct search identifies an early-visual advantage in NSD

The NSD search selects a fixed-weight encoder with a region-specific performance profile (Fig. 3). In search subjects 1 and 2, it exceeds every measured reference in V1, V2 and V3. The ordering reverses for the ventral aggregate, where pretrained representations are stronger. The comparison therefore supports a narrower claim than general model superiority: direct program search identifies a representation that is especially competitive for early cortical responses.

Subjects 5 and 7 show the same descriptive ordering even though neither contributes to program selection. The searched encoder remains the leading measured model in V1 through V3 and remains below the strongest pretrained reference in the ventral aggregate. The result provides evidence that the regional profile is not restricted to the two individuals used for search, while the small cohort does not support a population-level estimate.

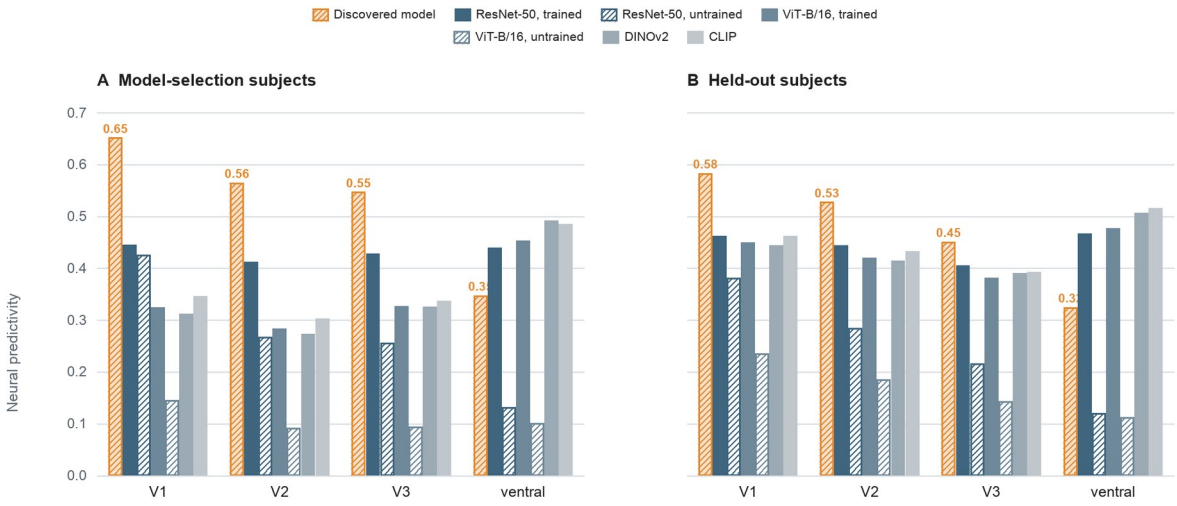


Figure 3 | Regional neural predictivity in NSD. Bars show noise-normalized explained variance for V1, V2, V3 and the ventral aggregate in the Natural Scenes Dataset. A, Mean performance across subjects 1 and 2, whose neural responses define model selection. B, Mean performance across subjects 5 and 7 after the selected program is frozen; these subjects do not contribute to program selection. The orange hatched bar is the searched fixed-weight program. Blue and gray bars are ResNet-50, ViT-B/16, DINOv2 and CLIP references; hatching distinguishes untrained encoders from solid pretrained encoders. Orange labels give the searched program's score. In both panels, the searched program leads the displayed references in V1 through V3 but not in the ventral aggregate.

Triple-N shows the same regional ordering

Triple-N changes both the biological preparation and the measurement scale. An independent search is driven by macaque single-unit responses rather than human fMRI. In the search cohort, the selected encoder leads the measured references in EVC but not in IT (Fig. 4). Because this is a separate program selected under a separate neural objective, the result concerns recurrence of a regional performance pattern, not transfer of the NSD model to macaque cortex.

The same ordering appears in the held-out cohort. EVC is available only in search monkey 1 and held-out monkey 4, whereas the IT bars average monkeys 1 to 3 for search and monkeys 4 and 5 for holdout.

The EVC comparison is therefore a pair of single-animal observations rather than a population estimate. Within that limitation, the searched model is again comparatively strongest in the earlier recorded region.

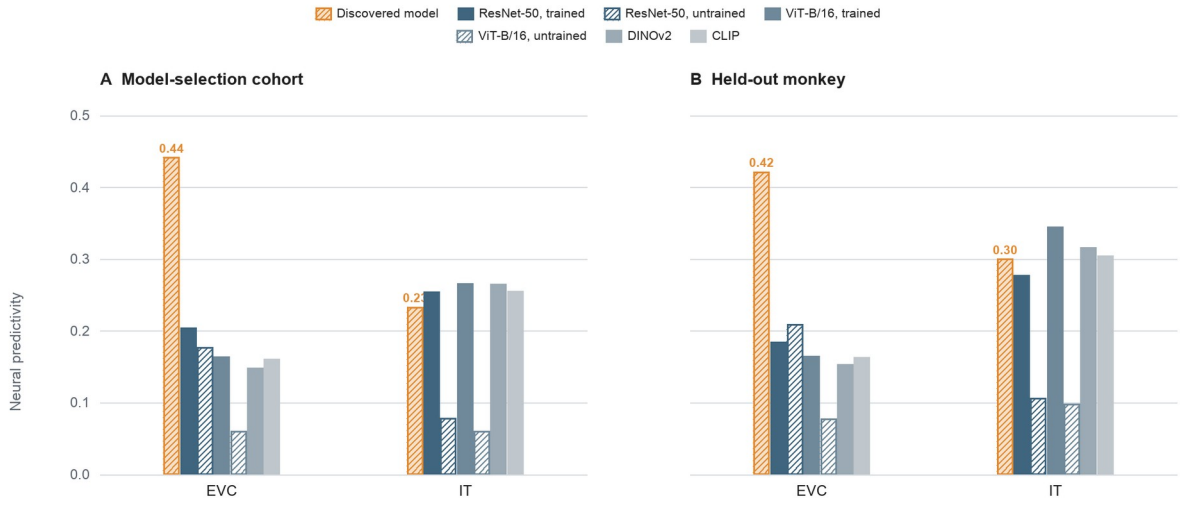


Figure 4 | Regional neural predictivity in Triple-N. Bars show noise-normalized neural predictivity for early visual cortex (EVC) and inferotemporal cortex (IT) in the Triple-N dataset. A, Search cohort comprising monkeys 1 to 3. B, Held-out cohort comprising monkeys 4 and 5 after the selected program is frozen. EVC is recorded only in monkey 1 for search and monkey 4 for holdout, whereas IT values average the corresponding cohorts. The orange hatched bar is the searched fixed-weight program. Blue and gray bars are trained and untrained reference encoders, with hatching identifying untrained models; orange labels give the searched program's score. The searched program leads the displayed references in EVC but not in IT in both the search and held-out cohorts.

TVSD extends the pattern to a distinct image and response regime

TVSD provides a third test with a different image collection, cortical sampling scheme and response metric. The model selected on monkey F leads the measured references in V1 and V4 but not in IT (Fig. 5). As in the other datasets, direct search does not produce uniformly higher predictivity across every recorded region.

Without further program selection, monkey N shows the same ordering across V1, V4 and IT. This single held-out animal provides a cross-animal check of the profile obtained in monkey F, not a population estimate. The result also shows that the descriptive pattern is not specific to the NSD or Triple-N stimulus and scoring protocols.

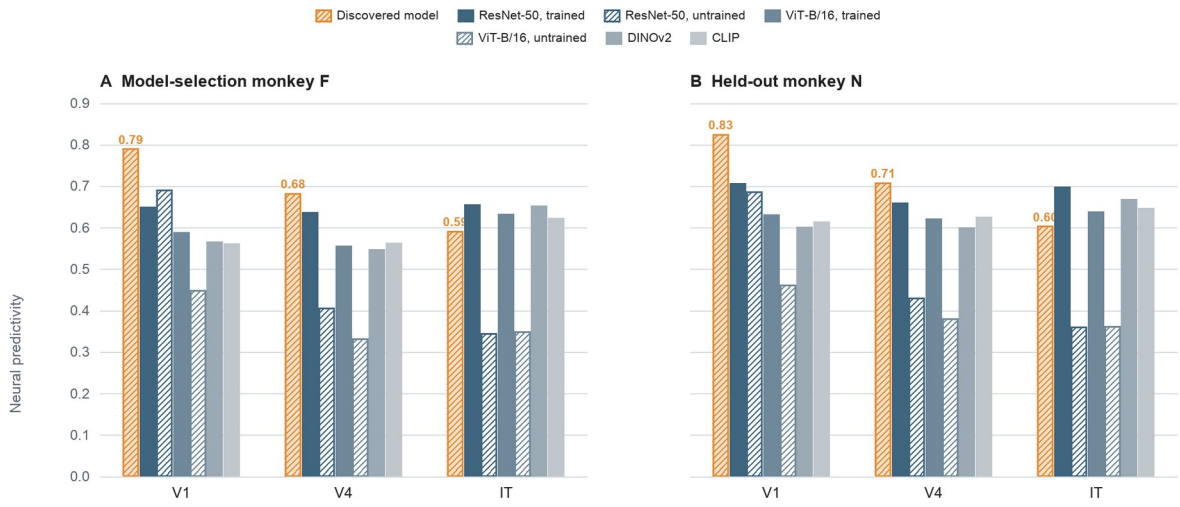


Figure 5 | Regional neural predictivity in TVSD. Bars show mean Pearson correlation between predicted and observed responses across reliable spiking units in V1, V4 and IT. A, Monkey F, whose responses define model selection. B, Held-out monkey N after the selected program is frozen; each panel therefore represents one animal. The orange hatched bar is the searched fixed-weight program. Blue and gray bars are trained and untrained reference encoders, with hatching identifying untrained models; orange labels give the searched program's score. The searched program leads the displayed references in V1 and V4 but not in IT in both animals.

Independent searches produce a shared regional pattern

The three searches select different programs under different neural objectives, and their scores cannot be pooled. What recurs is a coarse regional pattern: the searched programs are most competitive in earlier visual regions, while pretrained representations are stronger in later ventral or inferotemporal regions. The same within-dataset ordering appears in individuals excluded from program selection. This shared pattern marks the current empirical scope of the method; it does not imply that the programs share one implementation or transfer across datasets.

A preliminary look at what the discovered models compute

Our interpretation of the discovered models is still preliminary. We are currently developing more systematic analyses to determine which components drive neural predictivity, how the components interact and whether the same principles remain important across datasets. Here we make a narrower comparison based on the exported programs themselves: which broad computational ideas recur across the three independently selected models, and where do the programs appear to differ? This code-level comparison identifies operations that are available to the linear readout, but it does not establish that each operation is necessary or biologically implemented.

The clearest commonality is a progression from simple local measurements to more structured image descriptions. All three programs separate luminance and opponent-color information, measure oriented structure at multiple spatial scales, reduce sensitivity to local phase and normalize responses before spatial pooling. They then combine these signals to describe texture, contour and form. The exact combinations differ, but their role is similar: because the neural readout is linear, interactions between color and form, co-occurrences across orientations, or relations between curvature and position must already be explicit in the representation. The independent searches therefore converge on a broad design principle: normalized local oriented energy provides a common basis, followed by nonlinear combinations that make more complex visual structure accessible to a linear neural mapping.

The preliminary differences are easier to understand as emphases rather than as feature counts. The NSD program places greater emphasis on chromatic structure, foveal combinations and global image statistics. The Triple-N program gives more attention to log-polar geometry, curvature and the spatial arrangement of form. The TVSD program is more strongly organized around retinotopic oriented energy, junctions, end stopping and a fixed hierarchical form pathway. These observations do not yet support a human-versus-macaque distinction. The two macaque programs differ substantially from one another, and the searches also differ in their images, recording modalities, sampled regions, scoring rules and trajectories. For now, the strongest conclusion is the shared progression from normalized local energy to explicit form and texture combinations. Understanding which specific components account for each model's predictivity remains ongoing work.

Discussion

Computer vision has given neuroscience powerful models of natural images, but external engineering objectives need not define the full set of representations that neuroscience can test. Neuro-autoresearch complements pretrained models by using neural predictivity to order source-level programs. A selected program exposes its operations and fixed parameters directly, rather than locating most of its representational content in a learned checkpoint. This form makes candidate functions easy to rerun, inspect and modify within a fixed neural evaluation. Earlier models based on energy computations and hierarchical pooling show why explicit visual functions can be useful scientific objects.[19,20]

The results do not make pretrained networks unnecessary. They remain important references and are stronger in the later regions reported here. The early-versus-late pattern instead suggests a working hypothesis: fixed image operations may provide a sufficiently rich basis for some early visual responses, whereas later responses may depend more strongly on structure acquired through learning. Predictivity alone cannot distinguish a biologically relevant computation from a broad feature basis that a linear readout can exploit, so the present evidence does not identify a neural mechanism. Evidence that convolutional architectures can exhibit cortex-aligned representations before task training supports the broader importance of architectural bias, but it does not establish the mechanism of the programs selected here.[23]

The individual-holdout evidence is intentionally narrow. It tests whether a dataset-specific regional ordering is restricted to the individuals used for program selection. It does not establish direct transfer across datasets or species. The macaque comparisons also contain single-animal regional measurements, and the reference comparison is asymmetric because the searched program is selected from a large candidate population while each pretrained model is prespecified. The plotted differences should therefore be read as descriptive comparisons among the measured models rather than population-level estimates of superiority.

The programs nevertheless provide a more concrete object for interpretation than a score alone. Their named operations expose hypotheses about orientation energy, gain control, color opponency, texture and form. The present study establishes that these complete programs are predictive; it does not assign the observed performance to any one component. The distinction between an inspectable program and an interpreted mechanism is therefore central to the scientific claim.

The framework requires paired stimuli and neural responses, an executable stimulus-to-feature function and a held-out evaluation. That interface is not specific to vision, although no claim beyond the three visual datasets is established here. More broadly, open-ended program search gives computational

neuroscience a way to construct and test candidate representations directly, alongside the established practice of evaluating representations inherited from artificial systems.

Data availability

This study analyzes existing visual-neural datasets. Access to NSD, Triple-N and TVSD follows the terms of their originating projects. Source values underlying the reported figures are included with the project repository.

Code availability

The search framework, evaluation code and model definitions are available at github.com/yashsmehta/neuro-autoresearch.

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